

1. (1)
$$\begin{cases} \frac{1}{f} \frac{d^2 f}{dx^2} = -k_1^2 \\ \frac{1}{g} \frac{d^2 g}{dy^2} = -k_2^2 \\ k_1^2 + k_2^2 = 0 \end{cases}$$
 分3種情況:

(i) $k_1^2 > 0$:

$$\varphi(x, y) = (A_1 e^{k_1 x} + A_2 e^{-k_1 x}) (B_1 \sin(k_2 y) + B_2 \cos(k_2 y))$$

$$\varphi(x, 0) = 0 \Rightarrow B_2 = 0$$

$$\varphi(x, b) = U_0 \Rightarrow A_1 e^{k_1 x} + A_2 e^{-k_1 x} = \text{const} \neq 0, \text{ 矛盾}$$

(ii) $k_1^2 > 0$:

$$\varphi(x, y) = (A_1 \sin(k_1 x) + A_2 \cos(k_1 x)) (B_1 e^{k_2 y} + B_2 e^{-k_2 y})$$

$$\varphi(x, 0) = 0 \Rightarrow B_1 + B_2 = 0$$

$$\varphi(x, b) = U_0 \Rightarrow A_1 \sin(k_1 x) + A_2 \cos(k_1 x) = \text{const} \neq 0, \text{ 矛盾}$$

(iii) $k_1^2 = -k_2^2 = 0$

$$\varphi(x, y) = (A_1 + A_2 x) (B_1 + B_2 y)$$

$$\varphi(x, 0) = 0 \Rightarrow B_1 = 0$$

$$\varphi(x, b) = U_0 \Rightarrow A_1 + A_2 x = \frac{U_0}{B_2 b} = \text{const} \Rightarrow A_2 = 0$$

$$\Rightarrow \varphi(x, y) = A_1 B_2 y = \frac{U_0}{B_2 b} \cdot B_2 y = \frac{U_0}{b} y$$

(2) 令 $\varphi = \varphi_1 + \varphi_2$, 其中 $\varphi_1|_{x=0, 0 \leq y \leq b} = \frac{U_0}{b} y$, $\varphi_1|_{x>0, y=0} = 0$, $\varphi_1|_{x>0, y=b} = U_0$.

$$\varphi_2|_{x=0, 0 \leq y \leq b} = U_1, \quad \varphi_2|_{x>0, y=0} = \varphi_2|_{x>0, y=b} = 0$$

由(1)可知 $\varphi_1(x, y) = \frac{U_0}{b} y$

利用分離變量法, 設 $\varphi_2(x, y) = f(x)g(y)$

$$\begin{cases} \frac{1}{f} \frac{d^2 f}{dx^2} = -k_1^2 \\ \frac{1}{g} \frac{d^2 g}{dy^2} = -k_2^2 \\ k_1^2 + k_2^2 = 0 \end{cases}$$

結合 $\varphi_2|_{x=0, y=0} = \varphi_2|_{x>0, y=b}$ 的邊界條件, 考慮 $k_1^2 > 0$

$$\varphi_2(x, y) = (A_1 e^{k_1 x} + A_2 e^{-k_1 x}) (B_1 \sin(k_2 y) + B_2 \cos(k_2 y))$$

$$\varphi_2(x, 0) = 0 \Rightarrow B_2 = 0$$

$$\varphi_2(x, b) = 0 \Rightarrow k_2 = \frac{n\pi}{b}$$

$$\varphi_2|_{x \rightarrow \infty} = 0 \Rightarrow A_1 = 0$$

$$\varphi_2(x, y) = \sum_{n=1}^{\infty} C_n \exp\left(-\frac{n\pi}{b} x\right) \sin\left(\frac{n\pi}{b} y\right)$$

$$U_1 = \varphi_2(0, y)|_{0 \leq y \leq b} = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{b} y\right)$$

$$A_{1n} = \frac{2}{b} \int_0^b U_1 \sin\left(\frac{n\pi}{b} y\right) dy = \frac{2U_1}{n\pi} [1 - (-1)^n]$$

$$\Rightarrow \varphi_2(x, y) = \sum_{n=1}^{\infty} \frac{4U_1}{(2n-1)\pi} \exp\left(-\frac{(2n-1)\pi}{b} x\right) \sin\left(\frac{(2n-1)\pi}{b} y\right)$$

$$\varphi(x, y) = \varphi_1(x, y) + \varphi_2(x, y) = \frac{U_0}{b} y + \sum_{n=1}^{\infty} \frac{4U_1}{(2n-1)\pi} \exp\left(-\frac{(2n-1)\pi}{b} x\right) \sin\left(\frac{(2n-1)\pi}{b} y\right)$$

2. 通解: $\varphi(r, \theta) = \sum_{n=0}^{\infty} A_n r^n P_n(\cos \theta) + \sum_{n=0}^{\infty} B_n r^{-n} P_n(\cos \theta)$

邊界條件: $\varphi(a, \theta) = C \sin^2\left(\frac{\theta}{2}\right) = \frac{C}{2} (1 - \cos \theta) = \frac{C}{2} (P_0(\cos \theta) - P_1(\cos \theta))$

$$\begin{cases} A_0 + B_0 a^{-1} = -A_1 a - B_1 a^{-2} = \frac{C}{2} \\ A_n a^n + B_n a^{-n} = 0, n \geq 2 \end{cases}$$

$$\varphi(r, \theta) = \left[A_0 + \left(\frac{C}{2} - A_0\right) a r^{-1}\right] + \left[A_1 r - \left(\frac{C}{2} + A_1 a\right) a^2 r^{-2}\right] \cos \theta + \sum_{n=2}^{\infty} [A_n r^n - A_n a^{2n+1} r^{-(n+1)}] P_n(\cos \theta)$$

3. 不妨設 $\varphi(0, \theta) = 0$ [原點處(球心處)電勢為0]

在沒有 $\vec{E}_0$ 的情況下,

$$\varphi(r) = \begin{cases} \int_r^0 \frac{\rho}{3\epsilon_0} dr = -\frac{\rho}{6\epsilon_0} r^2, & r \leq R_0 \\ \int_r^R \frac{\rho}{3\epsilon_0 r^2} dr - \frac{\rho}{6\epsilon_0} R^2 = \frac{\rho R^3}{3\epsilon_0} \left(\frac{1}{r} - \frac{1}{R}\right) - \frac{\rho}{6\epsilon_0} R^2, & r > R_0 \end{cases}$$

在加了 $\vec{E}_0$ 的情況下,

$$\varphi(r, \theta) = \begin{cases} -\frac{\rho}{6\epsilon_0} r^2 + \sum_{n=0}^{\infty} (A_n r^n + A_{2n} r^{-n}) P_n(\cos \theta), & r \leq R_0 \\ \frac{\rho R^3}{3\epsilon_0} \left(\frac{1}{r} - \frac{1}{R_0}\right) - \frac{\rho}{6\epsilon_0} R^2 + \sum_{n=0}^{\infty} (B_n r^n + B_{2n} r^{-n}) P_n(\cos \theta), & r > R_0 \end{cases}$$

邊界條件: $\varphi(0, \theta) = 0$, $\vec{E}(r \rightarrow \infty) = \vec{E}_0$, $\vec{D}_{\text{外}}(\vec{R}_0) = \vec{D}_{\text{外}}(\vec{R}_0)$, $\varphi_{\text{外}}(R_0) = \varphi_{\text{外}}(R_0)$

$$\varphi(0, \theta) = 0 \Rightarrow A_{2n} = 0, A_{10} = 0$$

$$\vec{E}(r \rightarrow \infty) = \vec{E}_0 \Rightarrow \varphi(r \rightarrow \infty, \theta) \approx -E_0 r \cos \theta \Rightarrow \begin{cases} B_{10} = 0, n \geq 2 \\ B_{11} = -E_0 \end{cases}$$

所以 $\varphi(r, \theta) = \begin{cases} -\frac{\rho}{6\epsilon_0} r^2 + \sum_{n=1}^{\infty} A_n r^n P_n[\cos \theta], & r \leq R_0 \\ \frac{\rho R^3}{3\epsilon_0} \left(\frac{1}{r} - \frac{1}{R_0}\right) - \frac{\rho}{6\epsilon_0} R^2 + B_{10} - E_0 r \cos \theta + \sum_{n=0}^{\infty} B_{2n} r^{-n} P_n[\cos \theta], & r > R_0 \end{cases}$

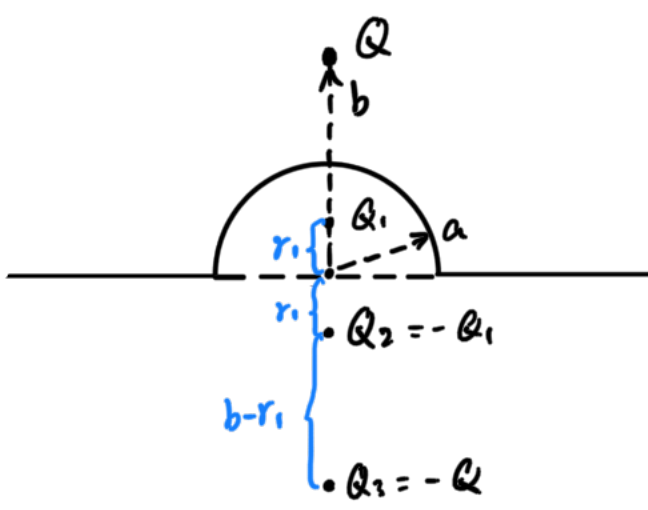
$$\begin{cases} \vec{D}_{\text{外}}(R_0) = \vec{D}_{\text{外}}(R_0) \\ \varphi_{\text{外}}(R_0) = \varphi_{\text{外}}(R_0) \end{cases} \Rightarrow \begin{cases} -\frac{\rho}{3} R_0 + \epsilon \sum_{n=1}^{\infty} n A_n R_0^{n-1} P_n[\cos \theta] = -\frac{\rho R_0}{3} - \epsilon_0 E_0 \cos \theta - \epsilon_0 \sum_{n=0}^{\infty} (n+1) B_{2n} R_0^{-n-1} P_n[\cos \theta] \\ -\frac{\rho}{6\epsilon_0} R_0^2 + \sum_{n=1}^{\infty} A_n R_0^n P_n[\cos \theta] = -\frac{\rho R_0^3}{6\epsilon_0} + B_{10} - E_0 R_0 \cos \theta + \sum_{n=0}^{\infty} B_{2n} R_0^{-n} P_n[\cos \theta] \end{cases}$$

$$\Rightarrow \begin{cases} \epsilon_0 B_{20} R_0^{-2} = 0 \Leftrightarrow B_{20} = 0 & (1) & \text{由(3)式和(5)式可知 } A_{1n} = B_{2n} = 0, n \geq 2 \\ \epsilon A_{11} = -\epsilon_0 E_0 - 2\epsilon_0 B_{21} R_0^{-3} & (2) & \text{由(1)式和(4)式可知 } B_{10} = B_{20} = 0 \\ \epsilon \times A_n R_0^{n-1} = -\epsilon_0 (n+1) B_{2n} R_0^{-n-1}, n \geq 2 & (3) & \text{由(2)式和(5)式可知} \\ B_{10} - B_{20} R_0^{-1} = 0 & (4) & -\epsilon_0 E_0 - 2\epsilon_0 B_{21} R_0^{-3} = \epsilon A_{11} = -\epsilon E_0 + \epsilon B_{21} R_0^{-3} \\ A_{11} R_0 = -E_0 R_0 + B_{21} R_0^{-2} & (5) & \Rightarrow (\epsilon - \epsilon_0) E_0 = (\epsilon + 2\epsilon_0) B_{21} R_0^{-3} \\ A_n R_0^n = B_{2n} R_0^{-n-1}, n \geq 2 & (6) & \Leftrightarrow B_{21} = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} E_0 R_0^3 \end{cases}$$

$$\Leftrightarrow A_{11} = -E_0 + \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} E_0 = -\frac{3\epsilon_0}{\epsilon + 2\epsilon_0} E_0$$

$$\varphi(r, \theta) = \begin{cases} -\frac{\rho}{6\epsilon_0} r^2 - \frac{3\epsilon_0}{\epsilon + 2\epsilon_0} E_0 r \cos \theta, & r \leq R_0 \\ \frac{\rho R_0^3}{3\epsilon_0} \left(\frac{1}{r} - \frac{1}{R_0}\right) - \frac{\rho}{6\epsilon_0} R_0^2 - E_0 r \cos \theta + \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} E_0 R_0^3 r^{-2} \cos \theta, & r > R_0 \end{cases}$$

5.



加入鏡像電荷 Q_1, Q_2, Q_3

其中 Q_1 是為了令 $r = a$ 的球面電勢 $\varphi = 0$

Q_2, Q_3 是為了與 Q_1 成折面對稱 $\Rightarrow$ 令平面電勢 $\varphi(r > a, 0) = 0$

$$r_1 = \frac{a^2}{b}, Q_1 = -\frac{Q}{b}, Q_2 = -Q_1 = \frac{Q}{b}, Q_3 = -Q$$

對於 $r > a, \theta \in (0, \frac{\pi}{2})$ 的區域,

$$\varphi(r) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{\sqrt{r^2 + b^2 - 2rb \cos \theta}} - \frac{Q}{\sqrt{r^2 + b^2 + 2rb \cos \theta}} - \frac{(a/b)Q}{\sqrt{r^2 + (a^2/b^2) - 2r(a^2/b) \cos \theta}} + \frac{(a/b)Q}{\sqrt{r^2 + (a^2/b^2) + 2r(a^2/b) \cos \theta}} \right)$$